

5(a)(i)	sum of electromotive force(s) = sum of potential difference(s) around a (closed) loop or the (algebraic) sum of the p.d.(s) and e.m.f.(s) is zero around a (closed) loop	B1
5(a)(ii)	(law of conservation of) energy	B1
5(b)(i)	(by Kirchhoff's first law) $I = I_1 + I_2$	B1
	$V/R_T = V/R_1 + V/R_2$ therefore $1/R_T = 1/R_1 + 1/R_2$	B1
5(b)(ii)	resistance of parallel combination = $(15 \times 10)/(15 + 10)$ (= 6.0 Ω)	C1
	$r = (E - V)/I$	C1
	$I = 1.38/6.0 = 0.23 \text{ A}$ $r = (1.50 - 1.38)/0.23$ $= 0.52 \Omega$	A1
	or	
	(by potential divider principle) $r/R_T = Ir/V$	(C1)
	$r/6.0 = 0.12/1.38$ $r = 0.52 \Omega$	(A1)
	or	
	(by potential divider equation) $V = E \times R_T/(R_T + r)$	(C1)
	$1.38 = 1.5 \times 6.0/(6.0 + r)$ $r = 0.52 \Omega$	(A1)

PUBLISHED

Question	Answer	Marks
5(c)(i)	as the (total) resistance has decreased (and e.m.f. is unchanged)	M1
	current will (in the cell) increase	A1
5(c)(ii)	(as greater current means a) bigger drop in p.d. across the internal resistance	M1
	p.d. (on voltmeter) will decrease	A1

Question	Answer	Marks
----------	--------	-------